Week 9

Case Study 1

Data for Questions 1 to 5

Customer Support Ticket Classification

A telecom company classifies customer support tickets into billing, network, plan-change, and device-support categories. After preprocessing 50,000 historical tickets, the team obtains a vocabulary of 12,000 unique words. As a baseline, every word is represented using a one-hot vector of length equal to the vocabulary size. Each vector has exactly one entry equal to 1 and all other entries equal to 0. A ticket is represented using a bag-of-words vector obtained by adding the one-hot vectors of the words present in it.

Consider two short processed tickets:

Ticket A: recharge failed refund

Ticket B: recharge plan refund

Both tickets are represented using bag-of-words vectors over the same 12,000-word vocabulary.

Question 1: What is the length of each one-hot word vector?

A) 12,000

B) 50,000

C) 3

D) 4

Correct Answer: A

Explanation: A one-hot word vector has length equal to the vocabulary size, which is 12,000.

Question 2: How many non-zero entries does a single one-hot word vector contain?

A) 12,000

B) 0

C) 1

D) 3

Correct Answer: C

Explanation: A one-hot vector has exactly one position set to 1.

Question 3: How many shared words appear in Ticket A and Ticket B?

A) 1

B) 3

C) 0

D) 2

Correct Answer: D

Explanation: The shared words are "recharge" and "refund", so there are 2 shared words.

Question 4: What is a major limitation of one-hot word vectors?

A) They are always dense for the data shown here.

B) They do not directly capture semantic similarity between words.

C) They require labels for every context window for the described model.

D) They have variable length for the same vocabulary.

Correct Answer: B

Explanation: Distinct one-hot vectors are orthogonal, so they do not encode semantic relatedness.

Question 5: Why can cosine similarity still be useful for bag-of-words ticket vectors?

A) It measures overlap in word usage while reducing sensitivity to length.

B) It learns neural hidden layers automatically from the sparse ticket vectors.

C) It converts sparse vectors into images in this specific scenario.

D) It guarantees semantic understanding under the given conditions.

Correct Answer: A

Explanation: Cosine similarity compares vector direction, which reflects word-overlap patterns in bag-of-words vectors.

Case Study 2

Data for Questions 6 to 10

Product Review Term Analysis

An online retailer studies product reviews using word co-occurrence statistics. The cleaned toy corpus contains four review fragments:

battery lasts long
screen looks bright
battery charges fast
bright screen response

The selected vocabulary is ordered as

$$[\text{battery}, \text{screen}, \text{bright}, \text{long}, \text{fast}].$$

The team uses a context window of 1, so only the immediate left and right words are counted as context. The co-occurrence matrix C has one row per target word and one column per context word.

Question 6: What is the size of the co-occurrence matrix for this selected vocabulary?

A) 4×5

B) 5×5

C) 5×4

D) 25×25

Correct Answer: B

Explanation: With five target words and the same five context words, the matrix has size 5×5.

Question 7: What does a context window of 1 mean?

A) Only the target word is counted.

B) All words in the corpus are counted as context.

C) The vocabulary is reduced to one word.

D) Only immediate neighboring words are counted.

Correct Answer: D

Explanation: Window size 1 uses the immediate left and right neighbors as context.

Question 8: In the fragment "battery lasts long", how many context contributions does "lasts" receive with window size 1?

A) 2

B) 1

C) 0

D) 3

Correct Answer: A

Explanation: The word "lasts" has "battery" on the left and "long" on the right.

Question 9: How many times are "screen" and "bright" adjacent in the toy corpus?

A) 0

B) 2

C) 1

D) 3

Correct Answer: C

Explanation: They are immediately adjacent only in "bright screen response", so the count is 1.

Question 10: What does a row of the co-occurrence matrix represent?

A) The number of labels in the classifier under the case assumptions.

B) The vectorial representation of a target word based on its context counts.

C) The one-hot position of a document category in the downstream product-review classifier.

D) The learning rate selected for truncated SVD.

Correct Answer: B

Explanation: Each row stores the target word's co-occurrence counts with context words.

Case Study 3

Data for Questions 11 to 15

Legal Search Term Representation

A legal search platform builds a word-context co-occurrence matrix from court-order summaries. The cleaned vocabulary has 6,000 words, so the square co-occurrence matrix X has size 6000×6000. The matrix is sparse and high-dimensional. To obtain compact dense word representations, the team applies truncated SVD:

$$X \approx U_k \Sigma_k V_k^\top,$$

with $k = 200$. The rows of $U_k\Sigma_k$ are used as dense word vectors for search and clustering. The platform wants related legal terms to be compared through shared context patterns rather than exact word overlap alone, which makes dimensionality reduction useful for retrieval.

Question 11: Why is truncated SVD applied here?

A) To increase the vocabulary size in the stated case setting.

B) To make every context vector one-hot.

C) To reduce dimensionality while preserving important structure.

D) To remove the need for context words in every downstream search task.

Correct Answer: C

Explanation: Truncated SVD gives compact dense vectors that retain important latent co-occurrence structure.

Question 12: What is the dimension of each reduced word vector when $k = 200$?

A) 6000

B) 36,000,000

C) 400

D) 200

Correct Answer: D

Explanation: The reduced representation has dimension $k = 200$.

Question 13: What is the size of U_k in this case?

A) 6000×200

B) 200×200

C) 200×6000

D) 6000×6000

Correct Answer: A

Explanation: U_k has one row per word and k columns, so its size is 6000×200.

Question 14: Which advantage does the reduced representation provide?

A) It makes all unrelated words identical under the given conditions.

B) It can make similarity computation more efficient.

C) It prevents any word from appearing in context.

D) It removes all latent semantic structure.

Correct Answer: B

Explanation: Comparing 200-dimensional dense vectors is more efficient than using full 6000-dimensional sparse rows.

Question 15: What kind of information can SVD help uncover?

A) The number of documents in chronological order.

B) The exact spelling of every rare word.

C) The chronological order of every document.

D) Latent patterns in word-context usage.

Correct Answer: D

Explanation: SVD can expose latent dimensions that summarize word-context usage patterns.

Case Study 4

Data for Questions 16 to 20

Search Term Similarity Evaluation

A technical support company compares two word-representation models for query expansion. For a set of word pairs, human annotators assign relatedness scores. The company computes cosine similarity between the learned word vectors and then measures how well model similarities correlate with human scores. The same vectors are also tested on a synonym selection task: given a query word and four candidates, the system chooses the candidate with the highest cosine similarity. For analogy checks of the form $a : b :: c :?$, the common search vector is $v_b - v_a + v_c$.

For one query word q, cosine similarities to four candidates are:

$$\text{A} : 0.18, \quad \text{B} : 0.74, \quad \text{C} : 0.41, \quad \text{D} : 0.62.$$

Question 16: Which candidate is selected by the synonym rule?

A) A

B) B

C) C

D) D

Correct Answer: B

Explanation: Candidate B has the largest cosine similarity, 0.74.

Question 17: What does a high correlation with human relatedness scores indicate?

A) The model's similarity judgments agree well with human judgments.

B) The vectors remain one-hot after training.

C) The vocabulary has only one word under the given conditions.

D) The model ignores cosine similarity during all human-score evaluations.

Correct Answer: A

Explanation: Higher correlation means model similarities align better with human relatedness ratings.

Question 18: In an analogy task $a : b :: c :?$, what vector operation is commonly used?

A) $v_a + v_b + v_c$ only

B) $v_a - v_b - v_c$

C) $v_b - v_a + v_c$

D) v_a / v_b

Correct Answer: C

Explanation: A common analogy query searches near $v_b - v_a + v_c$.

Question 19: Why are dense word vectors useful for query expansion?

A) They make every vector a one-hot vector under the given conditions.

B) They require no training corpus under the given conditions.

C) They make every pair of different words orthogonal.

D) They can place related words near each other in vector space.

Correct Answer: D

Explanation: Dense embeddings can encode distributional similarity, making related terms closer.

Question 20: What metric is used in the synonym rule described in the case?

A) Training loss under the given conditions

B) Matrix rank only

C) Cosine similarity

D) Vocabulary size

Correct Answer: C

Explanation: The case chooses the candidate with the highest cosine similarity.